The control limits for $\overline{X}$ chart is

$$UCL = \overline{\overline{X}} + A_2\overline{R} = 18.2 + 0.577(6.5) = 21.95$$
$$CL = \overline{\overline{X}} = 18.2$$
$$LCL = \overline{\overline{X}} - A_2\overline{R} = 18.2 - 0.577(6.5) = 14.5795$$

The control limits for Range chart is

$$UCL = D_4\overline{R} = 2.114(6.5) = 13.741$$
$$CL = \overline{R} = 6.5$$
$$LCL = D_3\overline{R} = 0(6.5) = 0$$

Example 9.22

The following data gives readings of 10 samples of size 6 each in the production of a certain product. Draw control chart for mean and range with its control limits.

Sample	1	2	3	4	5	6	7	8	9	10
Mean	383	508	505	582	557	337	514	614	707	753
Range	95	128	100	91	68	65	148	28	37	80

Solution:

Sample	1	2	3	4	5	6	7	8	9	10	**Total**
Mean	383	508	505	582	557	337	514	614	707	753	**5460**
Range	95	128	100	91	68	65	148	28	37	80	**840**

Table 9.22

$$\overline{\overline{X}} = \frac{\sum \overline{X}}{10} = \frac{5460}{10} = 546 \qquad \overline{R} = \frac{\sum R}{n} = \frac{840}{10} = 84$$

The control limits for $\overline{X}$ chart is

$$UCL = \overline{\overline{X}} + A_2\overline{R} = 546 + 0.483(84) = 586.57$$
$$CL = \overline{\overline{X}} = 546$$
$$LCL = \overline{\overline{X}} - A_2\overline{R} = 546 - 0.483(84) = 505.43$$

The control limits for Range chart is

$$UCL = D_4\overline{R} = 2.004(84) = 168.336$$
$$CL = \overline{R} = 84$$
$$LCL = D_3\overline{R} = 0(84) = 0$$

Example 9.23

You are given below the values of sample mean ($\overline{X}$) and the range (R) for ten samples of size 5 each. Draw mean chart and comment on the state of control of the process.

Sample number	1	2	3	4	5	6	7	8	9	10
$\overline{X}$	43	49	37	44	45	37	51	46	43	47
R	5	6	5	7	7	4	8	6	4	6

Given the following control chart constraint for : $n = 5$, $A_2 = 0.58$, $D_3 = 0$ and $D_4 = 2.115$

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